ETH Macro-Event Reaction Study — One-Page Research Summary

Asset/Data: ETHUSDT Binance USD-M perpetual, 1-minute klines, UTC. Event-day data spanning **2023–2026** (Powell/Jackson Hole, NFP, CPI, PPI, FOMC). Built and run in a cost-aware, leakage-controlled backtest framework.

MOTIVATION

The starting hypothesis was about **order flow around scheduled macro releases**: before a major announcement the market is balanced and uncertain; once the number lands and price makes a decisive move, flow may turn one-sided — so the initial breakout should *continue* rather than immediately reverse. To test continuation directly I built a **path-dependent dual-leg TP/SL bracket** on ETH perpetuals: simultaneous long and short with independent exits, so the winning leg can run while the losing leg is cut. (A *static* symmetric long+short on a linear perp is net-zero by construction — the bracket's entire payoff comes from asymmetric, path-dependent exits, i.e. it is a pure bet on continuation.) The bracket showed encouraging 2025 results but **no stable edge**: PPI whipsawed both legs, and performance did not replicate out-of-sample. Rather than tuning the bracket further, I used its failure pattern — clean one-sided breaks on CPI, whipsaws elsewhere — to ask a sharper question: is the reaction **directional**, not merely volatile? Isolating the long side produced the CPI pre-event long study below, and ultimately the regime-conditional conclusion.

FRAMEWORK & DATA DISCIPLINE

- Canonical UTC timestamp (`open_time_utc`); mandatory **Step-1 verification before every run**: checkpoints reconciled to raw 1m **100% on OHLCV**, `timestamp == event_time + offset`, 1440 candles/day, no duplicates. Runs **halt** on any missing event — never silently drop.
- Strategies fixed up front (pre-event long/short, post-event momentum, TP/SL straddle grid); no threshold tuning to fit results; fees/slippage modeled explicitly (0.05% fee + 0.05% slip).

EVENT SCREENING (2025)

Compared all event types on one footing and **screened out the noise**: - **PPI**: no tradeable reaction (whipsaw double-stops). **FOMC pre-long**: negative contributor. - **Ordinary Powell/Fed speeches**: nothing. **Jackson Hole**: large but a single event. - **CPI**: cleanest, broad-based reaction → became the focus.

2025 FINDING (IN-SAMPLE)

CPI pre-event long (enter −10m, exit +60m), post-May 2025, net of 0.2%/trade: **+9.5% cumulative, −1.08% max drawdown, 86% win (6/7)**. A direction-neutral **path-dependent TP/SL bracket** (TP +1.5% / SL −1.0% per leg): **11/11 positive after 4-leg fees**. *Note: a static symmetric long+short on a linear perp has no convexity and is net-zero; this positive result comes entirely from the path-dependent TP/SL execution, not from holding a true static straddle.*

MULTI-LAYER VALIDATION (THE CORE OF THE WORK)

Test	2025 (in-sample)	Out-of-sample / control	Verdict
Non-event-day baseline	CPI long mean at **97.5th pct** of random 60-min holds	random non-CPI holds avg **−0.16%**	**not pure beta**
CPI long, **2023–2024 OOS**	+9.5% (post-May)	**+0.06% mean, median −0.27%, 42% win**	does NOT replicate
CPI long, **2026 OOS**	+9.5% (post-May)	**+0.38% mean, 67% win (4/6)**	partial — weak-positive
Path-dependent TP/SL bracket, **2023–24 / 2026 OOS**	11/11 positive	**15/24 / 4/6**	does NOT fully replicate
Same-day **short** control	−11%	2023 −, 2024 −, **2026 −4.6% (0/6)**	**negative every year, 2023–2026**
Robustness across exit horizons	+15/+30/+60/+120m all positive	OOS all ≈0 / median negative	edge **not** an exit-time artifact

Regime gradient (the key pattern). CPI long directional return tracks the prevailing volatility regime monotonically: **2023–24 ≈0 → 2026 weak-positive (+0.38%) → 2025 strong (+1.31%/event)**. The effect strengthens with the regime rather than appearing/vanishing randomly — itself evidence the dependence is on regime, not on a single lucky year. *The post-May split was an ex-ante regime call (ETH entering a higher-volatility, higher- participation phase after reclaiming ~½ of its prior high), not a PnL-optimized cut — the 2023–24 and 2026 controls were run precisely to test, not flatter, that call.*

FINAL, FALSIFIABLE CONCLUSION

ETH's tradeable reaction to CPI — both the directional edge and the "always ≥1.5% breakout" volatility property — is **regime-conditional, not a stable structural property of ETH-vs-CPI.** The directional edge **scales with the volatility regime** (≈0 in 2023–24, weak in 2026, strong in 2025) and holds across all exit horizons, while the **short side is negative in every year 2023–2026** — so CPI drives a *directionally upward-biased reaction whose magnitude depends on the regime*, not two-sided noise. The headline 2025 result is a genuine regime effect, **not universal alpha** — established by explicit multi-year out-of-sample falsification rather than in-sample curve-fitting.

ANTICIPATED QUESTIONS & DEFENSE

- **"Isn't post-May 2025 cherry-picked?"** — Post-May was an *ex-ante regime call* (ETH entering a higher-volatility / higher-participation phase), **not** a PnL-optimized cut. The 2023–24 and 2026 out-of-sample windows were run to *test* that call; the result (effect is regime-dependent, not universal) is reported as-is.
- **"If a symmetric long+short is net-zero, why build a dual-leg bracket at all — and where does 11/11 come from?"** — No: the static symmetric long+short has **no convexity** and is net-zero by construction. The positive result comes from **path-dependent TP/SL execution** (asymmetric exits), not from holding a true static straddle. The bracket is deliberately *not* a static straddle — it is the direct test of the continuation hypothesis.
- **"Isn't the short-side result just crypto risk-on / long-run beta?"** — This **does not claim a stable standalone alpha**. It shows CPI reaction days carry a **directional asymmetry** relative to (a) random non-event 60-min holds and (b) same-day short controls — a *characterization of event behavior*, not an alpha claim.

SKILLS DEMONSTRATED

Event-study design; rigorous data integrity/verification; leakage control; cost/slippage modeling; alpha-vs-beta separation via baselines and directional controls; **out-of-sample falsification**; honest characterization of edge boundaries and single-event/regime dependence.